

Computer Forensics and Investigations: (01-14-26)

[Class One]

(All Canvas Assignments open).

PROF. MAX

Must attend Tests for attendance to be counted.

Digital Forensics is the application of comp. sci. and investigative procedures involving digital evidence analysis, after proper search authority, chain of custody, validation with mathematics, use of validated tools, repeatability, reporting, and possible expert presentation.

The Federal Rules of Evidence (FRE) was created to ensure consistency in federal proceedings. It was signed into law in 1973.

Investigating digital devices includes the following:

- Collecting data securely
- Examining suspect data to determine details such as origin and content
- Presenting digital info to courts
- Applying laws to digital device practices

Digital Forensics is different than data recovery, which involves retrieving information that was deleted by mistake or lost during a power surge or server crash.

you typically know what you're looking for in data recovery.

Forensics Investigators often work as part of a team, known as the **investigations triad**:

1. Vulnerability/Threat assessment and Risk Management.



2. Network Intrusion Detection and Incident Response

1. Tests and verifies the integrity of standalone workstations and network servers.

2. Detects intruder attacks by using automated tools and monitoring network email logs.

Digital investigations and conduction of forensic analysis of systems suspected of containing evidence.

A Brief History of Digital Forensics Tools

- By the early 1990s, the International Association of Computer Investigative Specialists introduced training on software for digital forensics.

- IRS created search-warrant programs

- ASR Data created Expert witness for Macintosh

- Ilook is currently maintained by the IRS Criminal Investigation Division

- Access Data Forensic Toolkit (FTK) is a popular commercial product

Understanding Case Law

- Existing laws can't keep up with the rate of technological change.

- When statutes don't exist, case law is used.
 - ↳ It allows legal counsel to apply previous similar cases to current ones in an effort to address ambiguity in laws.

- Examiners must be familiar with recent court rulings on search and seizure in an electronic environment.

Preparing for Digital Investigations

- Public sector investigations involve government agencies responsible for criminal investigations and prosecution.

- The Fourth Amendment restricts government search and seizure.

- The Department of Justice (DOJ) updates information on computer search and seizure regularly.

- Private sector investigations focus more on policy violations.

Understanding Public Sector Investigations

- When conducting public-sector investigations, you must understand laws on computer-related crimes including:

- Standard legal processes

- Guidelines on search and seizure

- How to build a criminal case

A criminal investigation usually begins when someone finds evidence of or witnesses a crime.

↳ A witness or victim makes an allegation.

A Digital Evidence First Responder (DEFr) arrives at an incident scene, assesses the situation, and takes precautions to acquire and preserve evidence.

A Digital Evidence Specialist (DES) has the skill to analyze the data and determine when another specialist should be called in to assist.

An affidavit is a sworn statement or support of facts or evidence of a crime.

↳ it must include exhibits supporting the allegation.

Understanding Private Sector Investigations

Private sector investigations involve private companies and lawyers who address company policy violations and litigation disputes.

↳ Ex: wrongful termination

Private Sector Crimes can involve:

email harassment, gender/age discrimination, falsification of data, embezzlement, sabotage, industrial espionage, etc.

- Most important policies define rules for using the company's computers and networks, known as an "Acceptable Use Policy".

A line of authority states who has the legal right to initiate an investigation, who can take possession of evidence, and who can access the evidence.

Business can avoid litigation by displaying it with warning banners on PC screens.

Businesses are advised to specify an authorized requester, who has the power to initiate investigations.

During Private Investigations, you search for evidence to support allegations of violations of a company's rules or an attack on its assets.

A private sector investigator's job is to minimize risk to the company.

Maintaining Professional Conduct

Professional conduct includes ethics, morals, and standards of behavior.

An investigator must exhibit the highest level of professional behavior at all times by maintaining objectivity and credibility/confidentiality.

Investigators should also attend training to stay with current changes.

Maintaining a Digital Forensics Investigation

The role of a digital forensics professional is to gather evidence to prove that a suspect committed a crime or violated a company policy.

Collect evidence that can be offered in court or at a corporate inquiry.

- investigate suspect's computer
- preserve evidence on a different computer

Chain of Custody is the route the evidence takes from the time you find it until the case is closed or goes to court.

Understanding Bitstream Copies

A bitstream copy is a bit-by-bit copy of an original storage medium, stored in a file called a bitstream image.